

Multi-Tenant Cost Allocation in Teams

Fair, transparent, and efficient AI infrastructure billing

A strategic approach to managing shared AI infrastructure costs across teams while maintaining fairness and efficiency



The Challenge of Shared Infrastructure

When multiple teams share the same AI cluster infrastructure:

Opaque Costs

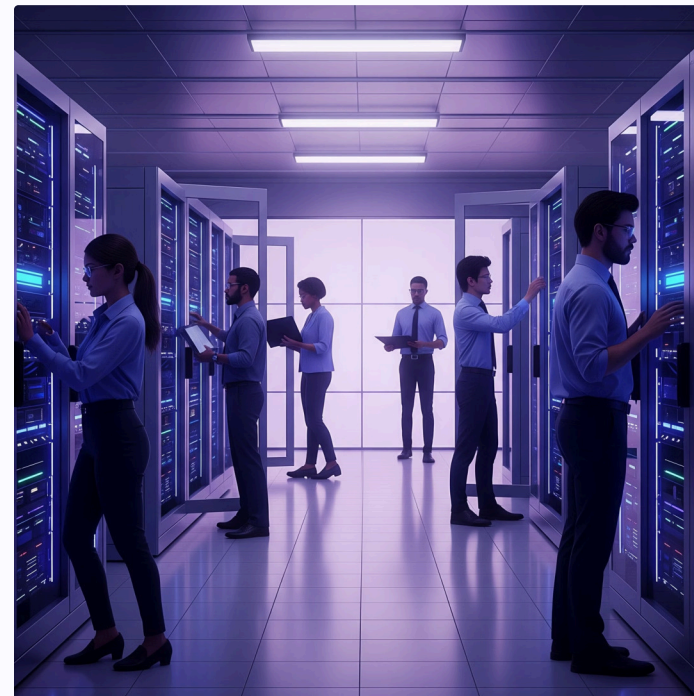
Without proper allocation, true resource consumption becomes invisible, making budgeting nearly impossible

Limited Accountability

Teams have little incentive to optimize resource usage when costs are distributed equally

Resource Contention

Critical AI workloads compete for limited GPU resources without clear prioritization



Our goal: Create a system that ensures fairness across workloads while maximizing infrastructure efficiency



Why Cost Allocation Matters

Prevents "Tragedy of the Commons"

Without accountability, shared GPU clusters become oversubscribed and inefficient as teams maximize their own benefit at collective expense

Financial Visibility

Finance teams gain granular understanding of AI investments, enabling better budgeting and forecasting for expensive AI infrastructure

Resource Optimization

When teams see their actual usage costs, they naturally implement efficiency improvements, reducing waste by 30-40% on average

Business Alignment

Enables chargeback/showback models that help align infrastructure costs with business value generated by different initiatives

Methods of Cost Allocation

1

Resource Tagging & Labeling

Implement consistent tagging across all infrastructure components:

- Team/department identifiers
- Project codes
- Application names
- Environment tags (dev/test/prod)

2

Quotas & Resource Limits

Prevent overuse through technical guardrails:

- GPU/CPU quota per namespace
- Storage limits by team
- Memory constraints
- Network bandwidth allocation

3

Usage-Based Metering

Track actual consumption metrics:

- GPU hours by workload
- Storage TB consumed
- Network egress
- Specialized hardware time

4

Cost Transparency Dashboards

Visualize spending patterns:

- Team-level spending trends
- Cost anomaly detection
- Utilization efficiency metrics
- Budget vs. actual comparisons

Tools & Platforms for Implementation



Kubernetes Native Controls

Resource Quotas, Limit Ranges, and RBAC to enforce team boundaries and prevent resource hogging



Kubecost

Purpose-built solution for K8s cost monitoring with team allocation, historical trends, and optimization recommendations



Cloud Provider Tools

AWS Cost Explorer, GCP Billing, and Azure Cost Management for native cloud resource allocation



Custom BI Solutions

Grafana, Tableau, or PowerBI dashboards integrating infrastructure and finance data for comprehensive views



The right combination of tools enables both technical and financial stakeholders to access relevant cost data in formats they understand

Implementation Challenges



Attributing Shared Resources

Network, storage, and control plane costs are difficult to divide accurately among teams using the same infrastructure



Workload Variation

ML training jobs, inference services, and data pipelines consume resources in fundamentally different patterns that simple metering can't equitably measure



Multi-Cloud Complexity

Different cloud providers use inconsistent billing models, making unified visibility across hybrid environments particularly challenging



Cost-Benefit Balance

Overly complex tracking systems can create more overhead than the savings they generate—finding the right granularity is critical



Best Practices for Success



"The most successful implementations start with visibility before accountability, focusing on education rather than punishment."

1 Align with Organizational Structure

Map cost allocation to how your company actually works—by team, project, product line, or business unit

2 Start with Showback, Then Chargeback

Begin by showing teams their costs without financial consequences, then gradually implement actual billing as behaviors adapt

3 Automate From Day One

Build tagging and monitoring into your CI/CD pipelines and infrastructure-as-code to ensure consistent application

4 Create Shared Incentives

Distribute a portion of cost savings back to teams that optimize their resource usage to encourage continuous improvement

Key Takeaways

Multi-tenant cost allocation is **essential** for scaling AI infrastructure effectively in enterprise environments.

Visibility drives efficiency – Teams optimize what they can measure, typically reducing waste by 30-40%

Practical tools exist – Kubecost, cloud billing tools, and custom dashboards provide actionable insights

Cultural transformation – Beyond tools, successful implementation creates a culture of financial responsibility



Start small, focus on high-value resources like GPUs, and build a system that evolves with your organization's needs

i Need help getting started? Contact your cloud architect or finance business partner to begin mapping your current infrastructure spending patterns.